

```
#include <ctype.h>
#include <stdio.h>

int tempCount = 0;

void generateCode(char op, char operand1, char operand2);
void intermediateCode(char *expr);

int main() {
    char expr[100];
    printf("Enter an expression: ");
    scanf("%s", expr);

    intermediateCode(expr);

    return 0;
}

void generateCode(char op, char operand1, char operand2) {
    printf("t%d = %c %c %c\n", tempCount, operand1, op, operand2);
    tempCount++;
}

void intermediateCode(char *expr) {
    char operandStack[100];
    int top = -1;

    for (int i = 0; expr[i] != '\0'; i++) {
        if (isalnum(expr[i])) {
            operandStack[++top] = expr[i];
        } else if (expr[i] == '+' || expr[i] == '-' || expr[i] == '*' ||
            expr[i] == '/') {
            char operand2 = operandStack[top--];
            char operand1 = operandStack[top--];

            generateCode(expr[i], operand1, operand2);
            operandStack[++top] = 't';
        }
    }
}
```